For MR750:

Channel	*Unblank signal*	*UPM block*
HEAD	NB	1
BODY	NB	2
BB	BB	3
CW	CW	4

For MR750w:

Channel	*Unblank signal*	*UPM block*
HEAD	NB	1
BODY CH1	NB	2
BB	BB	3
BODY CH2	NB	4

In MR750w, since UPM block 4 monitors BODY CH2 transmit channel, it also needs to use the NB unblank signal (as opposed to the CW unblank signal as in MR750). This change was achieved by modifying the unblank cable feeding the UPM. The NB unblank signal was hard-wired to drive both the NB and CW inputs to the UPM. Therefore, from the power monitor's point of view, in 750w, the BODY CH1 and BODY CH2 unblank signals are both derived from NB unblank, and are always in lockstep with each other.

This cable change also limited the scope of the change. Other components that receive the "CW unblank" signal remain unaffected because the cabling change affects only the unblank signals provided to the UPM.

4.7.6.2 XMIT mode

During TRANSMIT, the RF amplifier is enabled (put in a ready to transmit state) and unblanked (gated on), receive chain transmit protection is enabled, and transmit blocking or protection for non-transmit RF coils is enabled. More details on TRANSMIT:

RF amplifiers: When RF amplifiers are "unblanked" they are ready to amplify input signals to high power RF. The solid state amplifiers used in DV require 50µsec, after the UNBLANK leading edge, to enable and stabilize the high power stage. Therefore, pulse sequences must incorporate a 50µsec delay after the unblank before commanding the exciter to produce RF.

Receive chain protection: The transmit chain provides the transmit protection to the receive chain components, primarily the preamplifiers. The transmit pulses are on the order of 1000V; preamplifiers may be damaged by 10V or so. To protect the preamps, *TR switches* are designed break open the pathway between the transmit path and the receive path to the preamp. Typical TR switch isolation values are on the order of 60-65 dB. This isolation is achieved using PIN diode switches; the bias used to control these switches is called *TR bias*. The driver module creates TR bias signals using the NB (and BB) unblank signals as control. . The driver module requires that either blank or unblank be maintained for at least 200µsec. This places a limitation pulse sequence design, as each state must be at least ≥200µsec long, even if that time is not necessary for other components (like the RF amplifier) or the pulse width, as in the Silent sequences.

MC bias: In DV the RF Hub contains the circuitry that creates the multi-coil (MC) bias necessary to protect the non-transmit (or receive-only) surface coils. The Hub receives the same unblank control signals used by the driver to synchronize the bias with the sequence. The MC drivers in the Hub are capable of switching times of 20µsec or less. This means that the drivers in the Hub can adequately switch, settle and monitor the outputs for fault conditions in 20µsec.

Dynamic disable drivers: The unblank signals used by the transmit chain are also used to control the Dynamic Disable (DD) drivers when the system is scanning in surface mode. The system body coil can be enabled throughout the entire scan as when the body coil acts as both the transmitter and receiver coil (body mode),

disabled throughout the entire scan as when a local TR coil is used that acts as the transmitter and receiver coil (head mode) or it can toggle between the enabled and disabled state as when it acts as the transmit coil but the receive coil is the surface coil (surface mode). When the body coil must toggle between the enabled state and disabled state, the driver module uses the unblank signal to control the DD drivers that provide the proper bias to the PIN diode circuits during the XMIT state and RCV state.

4.7.6.3 RCV mode

During RECEIVE, RF amplifiers are blanked or disabled, the receive chain is enabled, and non-transmit RF coils are enabled. More detail on RECEIVE is below:

RF amplifiers: When the RF amplifier is blanked, it must 1) prevent any signal from passing thru the amplifier, and 2) have low output noise. In the blanked state, even if an RF input is present at the amplifier's input, there will be no output; an internal switch is opened in the signal path. The amplifier's output noise is guaranteed to be within 20dB of the -174dBm/Hz thermal noise floor within 10µsec of activating the blank signal.

Power monitoring: As a safety mechanism, the power monitor will establish a fault condition (which is used to immediately stop any scan in progress) if RF output power is sensed during the blank state.

Receiver enabling: During RECEIVE, the receivers are enabled, and can acquire data – a bit confusingly, we also use the term "unblank" to apply to receiver enabling. (During TRANSMIT, the receivers are "blanked" to protect the front end components from saturation effects due to the large RF transmit pulse.)

TR and MC drivers: During receive the TR drivers change state such that the PIN diodes in the TR Switches are back-biased and create a low-loss path from the receive coil to the preamplifier. The MC drivers in the RF Hub change state to enable the surface coil elements to pick up the MR signal from the body.

4.7.7 RF Power Monitor

RF transmit power is monitored to protect the patient from excessive energy deposition. MR750, MR450, MR450w, and MR750w all use the Universal Power Monitor (the UPM).

A UPM consists of one or two RF Detector Board(s) – the RFDBs or just DBs – and one Processor Board (UPB). Roughly speaking, the DBs handle analog signal processing, and the UPM handles digital signal processing. The DB accepts sampled RF signals, detects the RF envelope, logarithmically scales it, and digitizes it. The UPB takes the digitized RF signal from the DB, analyzes the RF power and calculates SAR. It also serves as the communication connection to the system. All intelligence, analysis and fault detection occurs on the UPB.

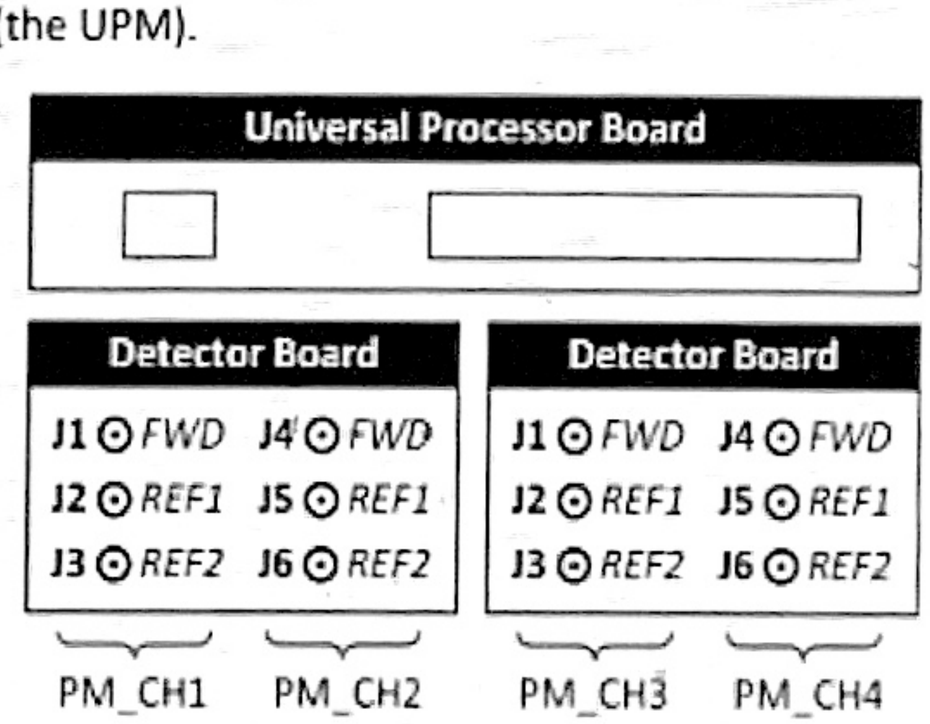


Fig. 4–28: The Universal Power Monitor

A UPM with one detector board has two "channels", PM_CH1 and PM_CH2. A two DB UPM has four channels, adding PM_CH3 and PM_CH4. Each channel is typically used to monitor one power amplifier, and has inputs for three RF signals. See §4.7.7.1 for more details.

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For compatibility with both legacy and current systems, the UPM can operate in two modes. In all DV systems, the UPM operates in *overage mode* where net power to the patient is averaged over two time windows: a 10 second "short term" and six minute "long term" window. (On the 1.5T HD platform, the UPM uses *pulse characterization mode* where pulse width, duty cycle and pulse amplitude are monitored. This mode will not be covered in this document.)

All systems contain two identical but independent UPMs for redundancy, designated as UPM1 and UPM2. Therefore, at a minimum, each system will contain two Processor Boards and two Detector Boards, as well as two sets of sample cables for each utilized sample port. The UPM indicating the higher power level is always used for fault detection.

4.7.7.1 RF Detector Board

At a high level, the function of the RFDB is to accept coupled RF power samples from the system transmit hardware (usually, from directional couplers inside the RF amplifiers) and convert the coupled signals to digitized waveforms representing the modulation envelopes of the RF pulses. The DB is broadband, so it can be used for 1.5T (64 MHz) systems, 3T (128 MHz) systems, and MNS (16-130 MHz).

The original Detector Board design provided six separate sample inputs, divided into two channels, with three inputs per channel. Each input can process one RF signal. The six circuit paths are independent and identical such that they could be, in theory, interchangeable.

The convention is to use the three inputs as FWD (forward transmit power from the amplifier), REF1 (power reflected to the amplifier) and REF2 (power reflected to a dummy load by a quad splitter).

In systems that do not require reflected power samples, a depopulated DB can be used to reduce cost.

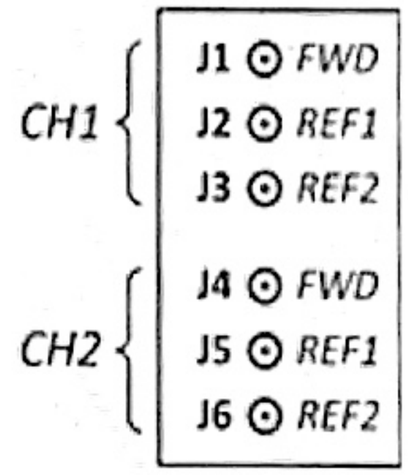


Fig. 4–29: The RFDB

The core signal processing for each input of the DB is shown below. There are four functions: cable detect, signal scale, logarithmic detection, and A/D conversion.

Cable detect: Since the UPM serves a critical safety related function, foreseeable faults must be prevented from affecting its function. A disconnected cable is detected, since the DBs look for a 50Ω source impedance, and create a fault condition if this impedance is not detected. (Therefore, all unused ports must use 50Ω terminations.)

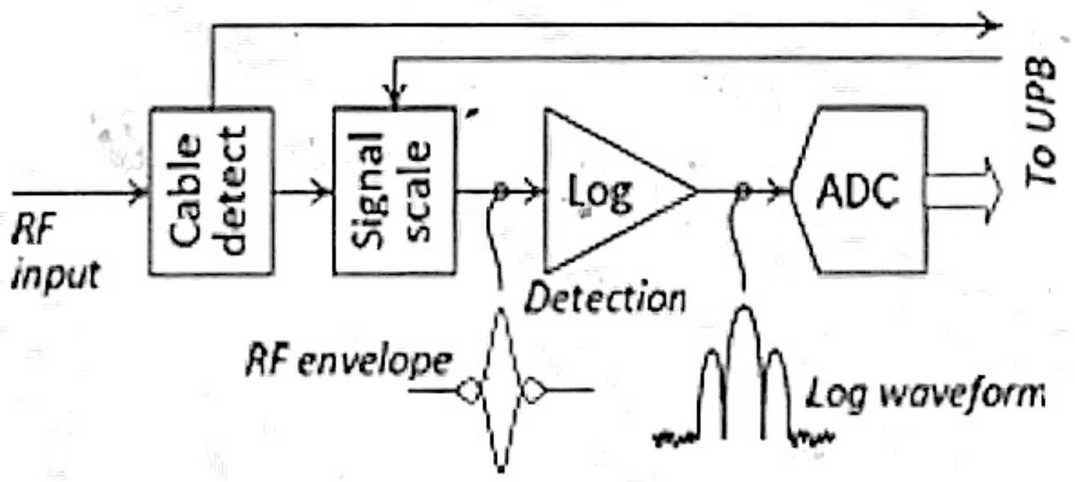


Fig. 4–30: RFDB block diagram

Signal scale: The RFDBs have the capability to normalize their input signals with a calibration routine. This feature provides two benefits. First, it allows the DB to compensate for amplifier coupler and/or cable attenuation variations, which improves the overall UPM accuracy. Second, since the DB outputs are adjusted to a consistent level, it allows for use of a single conversion tables in the UPB.

During calibration a signal of known amplitude is input to the DB, and internal adjustments are made to normalize the sample's peak level such that the ADC performance is optimized. The calibration routine produces calibration values, which are stored in files `/w/config/Upm1Cal.cfg` and `/w/config/Upm2Cal.cfg`

Logarithmic detection: The RFDBs use a logarithmic detector, as opposed to a linear detector. This allows the DB to handle a very large range of input signal amplitudes, and detect very low signal levels. The digitized waveform sent to the UPB follows the log of the waveform, rather than the waveform itself.

A/D conversion: The final stage is A/D conversion. The original UPM implementation used 100KHz sampling; this was increased to 1MHz sampling for compatibility with "silent" imaging (see §4.7.7.5).

4.7.7.2 Universal Processor Board

The UPB provides signal processing for the RF envelope output of the DBs: filtering and averaging these signals and comparing them to limits downloaded from the system. If the limits are exceeded, the UPB generates a fault condition, sends it to the system, and inhibits an amplifier unblank signal, effectively stopping the RF amplifier from producing RF power.

The UPB also provides control and communication for the UPM. The UPB communicates with the system via the CAN (Controller Area Network) network.

The UPB uses the data in the calibration files along with the inputs from the Detector Board ports to determine the power level present on each sample. The UPB will prevent scanning (except for geservice scanning) if the calibration files do not contain valid data. It is responsible for maintaining a short and long term average power level for each of three specified anatomies (head, body and extremity).

The Processor Board relies on information in configuration file `/w/config/UpmConfig.cfg` file to translate the data from the RFDB into actual power. This file contains information on the scaling factors of the selected RF amplifiers for the system.

4.7.7.3 DV System UPM Configurations

The UPM's multiple channel design can be used with a wide range of transmit configurations. The table below summarizes the configurations used on the MR450/450w/750/750w systems.

System type	*PM_CH1*	*PM_CH2*	*PM_CH3*	*PM_CH4*
MR750w	Body_CH1	Head	Body_CH2	MNS_BB
MR750	Body	Head	??	MNS_BB
MR450/MR450w	Body	Head	N/A	N/A

While the DBs are all identical, needing only recalibration to adapt of different configurations, the UPM is not as flexible. Since the UPM has body, head, and MNS paths mapped in its internal firmware, the PM_CH# assignments for body, head and MNS samples must remain as shown above.

Average vs. local SAR (included in the patient safety section) - for information on SAR and SAR control see section 4.2.2 (Patient Safety)

The UPM is directly involved in measuring and calculating the amount of whole body SAR. Local SAR is controlled using models and an understanding of the relationship between local SAR and whole body SAR.

All DV systems monitor power in Average Mode using a short (10 second) and long (6 minute) term moving window. Measured SAR (Specific Absorption Rate) is calculated using the net power delivered to the patient. This net power is a function of measured forward power, measured reflected power as well as a statistical energy transfer efficiency factor that includes system losses in the coil and cables. In systems where both forward and reflected powers are measured, the net power is derived directly from the two measurements. In systems where only forward power is measured, the fraction of power lost due to reflection is based on a system model that is a function of patient weight.

4.7.7.4 Time domain filtering

There is a "several-orders-of-magnitude" difference in the bandwidth of a) the RF pulses used in MR and b) the body's response to SAR. RF pulses are on the order of milliseconds, while the thermal response to SAR is over many seconds or even minutes. Therefore, the UPM applies time-domain filtering to emulate the body's response to RF absorption.

The time domain filtering is done is several stages:

1. First, the RF envelope samples are summed up into 100ms long blocks. This is not really a filter, in the conventional sense, as much as an integrate-and-reset operator. The output is one point every 100ms, that equals the total energy in the previous 100ms. These 100ms samples are multiplied by a scaling factor to convert energy to SAR.
2. Second, the 100ms "SAR" samples are run thru a boxcar filter (aka moving average), with a 10s window width. The output is updated every 100ms, and equals the SAR, averaged over the last 10 seconds.
3. Third, another boxcar filter – this time with a 6 minute window width – provides the SAR averaged over the last 6 minutes. This filter picks up every 10th sample from the 100ms filter – which is one sample every second. The last six minutes worth of data (600 samples) are averaged to form the output of the filter.

The average SAR outputs of the 10s and 6m boxcar filters are compared to limits (calculated by the system and downloaded to the UPM). If either filtered output exceeds the matching limits, a fault is generated and and scanning is inhibited.

4.7.7.5 Silent imaging compatibility

The original UPM was designed for a maximum RF pulse bandwidth of approximately 40kHz, and used a digitization rate of 10µs. This accurately captured the RF pulses used on pre-2012 systems, where pulses used bandwidth limited shapes, and that were at least several hundred microseconds long.

The Silent imaging technique (introduced in 2013 with DV24) changed pulse timing by (roughly) two orders of magnitude. Silent uses very short, hard edged rectangular pulses (as sort as 8µs) and reduced the allowable time between the end of the RF pulse and the end of amplifier unblanking (to roughly 20µs). This forced a UPM design update, to handle the high RF envelope bandwidth, short pulse width and short time delay between the end of the RF pulse and the blank portion of the sequence.

To address these issues, several changes were implemented.

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1. The RFDB was re-designed to optimize the log detector's transient response and filtering. The log detector's low pass filter bandwidth was increased to approximately 450 kHz, allowing the detector to accurately follow the high slew rate input signal. In addition, the end-of-pulse response was optimized to minimize the detector's "ringdown" (the long decay tail that follows the abrupt end of a pulse). This reduced overlap between the ringdown tail and end of unblank, which would cause the UPM to generate a "power during blank" fault.
2. The UPB's assembly and firmware code was updated for two main reasons. First, the sampling interval decreased from 10µs to 1µs to handle the higher pulse bandwidths; this required a larger FPGA to handle the higher data rates associated with the 1µs sampling. Second, the "power during blank" fault detection strategy needed to be updated to tolerate the post-pulse decay (aka "logamp ringdown") inherent in the log detectors. The fault detection logic was changed so several consecutive data points above a detection threshold were needed to cause a fault.

Although these changes were made for compatibility with the new pulse sequence, the changes allow backwards compatibility; the updated UPM is compatible with all scanning and maintains the accuracy necessary for the power monitoring scheme.

4.7.7.6 Multiple transmit ports

The DV system has three separate transmit paths with the capability of providing RF power into coils in the scan room. These three paths are designated body, head (or local transmit) and broadband MNS (optional). The Multi-Nuclear Spectroscopy information will not be discussed here. All DV systems have the capability of transmitting narrowband body power into the body coil and narrowband head power into T/R coils connected at the A-connector or P1-connector. Therefore all DV systems are configured to monitor power on the body path and the head path. RF power on any path must be coincident with the appropriate unblank signal. In other words, if power is detected on either the head or body path and the narrowband unblank signal is not asserted, a fault will be flagged. Since the UPM has the capability of monitoring all ports simultaneously, it can also detect if power is being transmitted on the "wrong port" even if the correct unblank signal is available (such as body verses head).

In Dual Drive systems the UPM has the capability of monitoring both body channels simultaneously as shown above. Originally the Body CH2 ports were designed to monitor a narrowband CW amplifier that may have been incorporated in future systems. However, since the DV amplifiers all have CW capability (duty cycle = 100%), this additional amplifier became unnecessary. Therefore when the need arose to monitor a second body channel, these UPM ports were re-purposed with no change needed to the hardware.

4.7.7.7 Allocation of power to anatomy

Since not all coils deposit power to the patient at the same spot at the same rates, the SAR measurement is divided up into three different meters associated with different areas of the anatomy. These three meters are designated whole body, head average and extremity average. The power monitoring system is responsible for mapping how the transmitted power is distributed in the patient using the information it has available about the transmit coil and the patient. The limits for the head and body meter is made available to the power monitor by the host and can be found in a configuration file in the /w/config directory called Scansar.cfg. The limit for the extremity meter is calculated based on a partial body mass model. The short

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term SAR limits are set to a constant scale factor times the associated long term SAR limit. This constant is currently set to 3.0. Information on the transmit coil characteristics that affect the SAR limit is found in the coil database. Note that although these allocations based on anatomy have similar names to the physical sample ports identified with the UPM, they are not the same concept. The physical port on the UPM designated "head" is actually measuring power on the cable paths connecting the amplifier to the local transmit coil connected to the system. The meter designated as "head" is the amount of power expected to be dissipated in the head-no matter which transmit path is providing the power. For example, depending on the anatomy being scanned, power from the body output port of the amplifier can affect the amount of power in the "head" meter.

4.8 Receive chain design

The receive chain, or *RXC*, contains the components used to process the received MR signals. This includes 1) preamplifiers, 2) demodulation and digitization signal paths, and 3) time-based filtering. Generally, the RXC begins at the preamplifier inputs, and ends with the digitized time-domain data in the rawdata files.

In addition, the RXC includes the user accessible RF coil connection ports: the "A" and "P" type ports.

Considerations include

1) Dynamic range
2) Bandwidth
3) Noise figure
4) Maximum signal handling
5) Transmit signal handling
6) IM distortion
7) Single tone purity
8) T/R switching time
9) Multi-channel crosstalk
10) Multi-channel gain and phase balance
11) Transient noise suppression
12) Receive signal routing
13) Receive signal processing
14) Frequency conversion

4.8.1 Subsystem block diagram

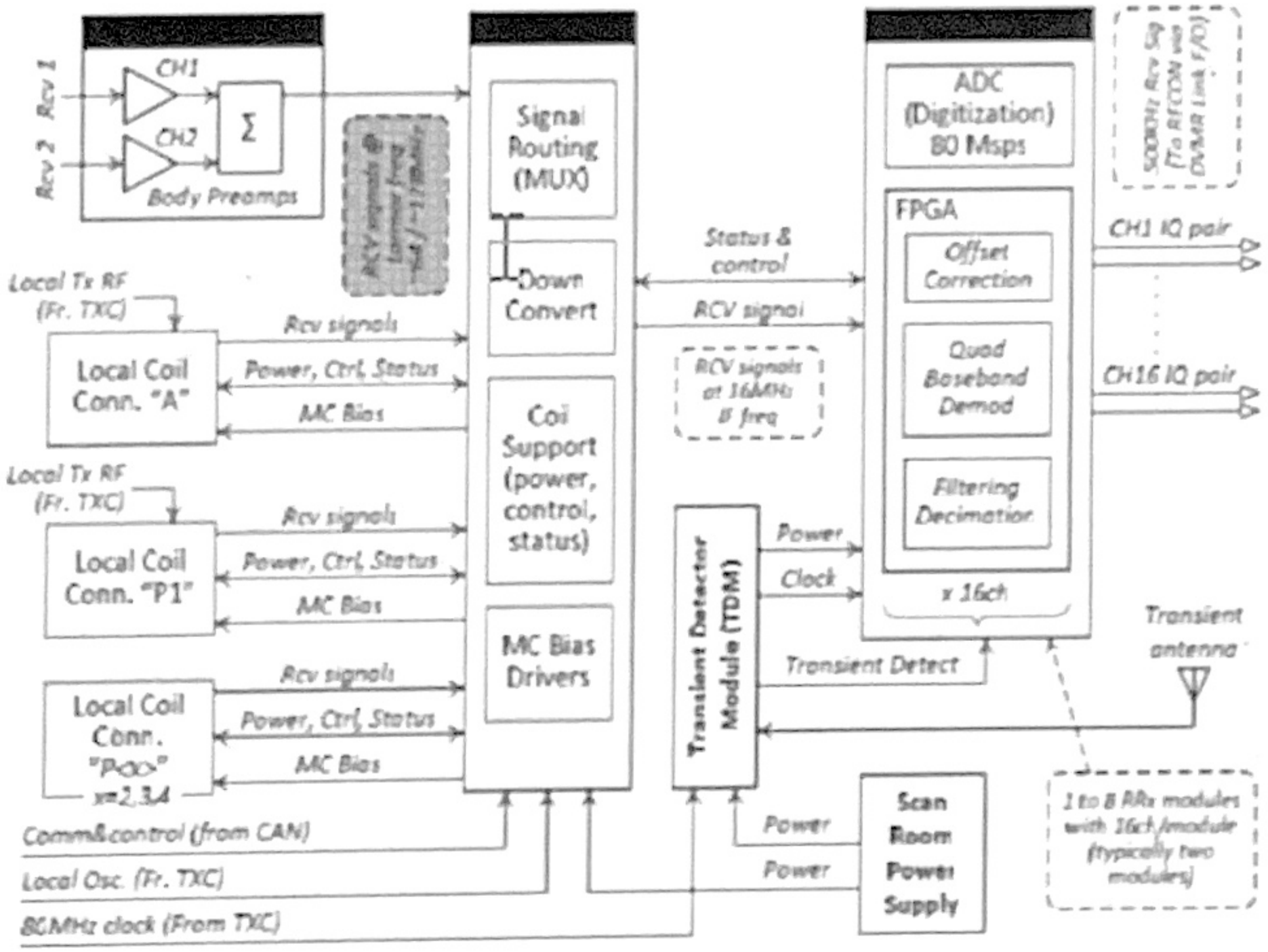


Fig. 4–31: RXC block diagram

Local coil interface

All DV-compatible receive-only surface coils have 1) onboard preamplifiers, and 2) a means to provide coil identification, including the coil type and unique serial number. The surface coil interfaces are multiple pin connections that allow for quick disconnect capability, used if it is necessary to remove the patient from the system in an emergency situation.

Two of the connectors, designated Connector A and P1, have transmit capability for local transmit coils (T/R Coils).

The receive chain provides power (for coil components such as preamps) and control signals (for functions such as on-coil MUXing of elements) and receives status information from the coil through this connection. The RxC also provides decoupling bias for multi-coil arrays that provide protection for the surface coil during body coil transmit. The coil decoupling networks prevent or reduce circulating currents in the coil loops during transmit. Circulating currents can cause inhomogeneity in the B1 field (which would degrade the image) as well as cause coil heating (which can be a patient safety concern).

Body coil interface

The body coil receive path uses two preamplifiers and a combiner circuit to produce a single receive signal. The main function of the preamp/combiner is to provide low-noise amplification for the received signal, so

body preamp noise figure is a critical parameter. The power for the preamps travels on the same coaxial cable that is used for the RF receive signal.

The RF Hub

The RF Hub resides in the rear pedestal. Its main function signal routing between the coil ports and receivers. It also contains the circuitry for multi-coil bias (including fault monitoring), the communication link to the system (CAN) and manages control and status to and from the coil ports.

The CAN interface is used only when the system is not scanning because the DSP used for this communication protocol can generate noise that can degrade the image. This communication interface is used to pre-configure the Hub appropriately before the scan (i.e. set up the receive paths and multi-coil driver configurations). Any real-time control needed during the scan comes to the Hub through the Receiver. The Receiver gets this real-time information via the DVMR Link.

The Hub also contains down-conversion circuits to translate the RF signal from the Larmor Frequency (64 MHz or 128 MHz) to the Intermediate Frequency (IF) of 16 MHz.

In 1.5T the LO frequency is 80 MHz and in 3T it is 112 MHz. The LO is generated in the DV Exciter and is sent to the RF Hub via a twinax cable as a differential signal to minimize noise susceptibility.

Receivers

The DV Receiver contains the Analog to Digital Converter (ADC) used to digitize the signals and a Field Programmable Gate Array (FPGA) where the demodulation, filtering and decimation take place. The RXC performs a calibration to measure and characterize any ADC offsets; if this offset is not removed, it may cause image artifacts. The FPGA applies the appropriate offset correction to the signal. The ADC uses the system clock and therefore operates at 80 Msamples/sec.

The system clock is generated by the DV Exciter and is sent to the Receiver(s) via the Transient Detection Module (TDM) on a twinax cable. This signal is transferred differentially for noise immunity. Each receiver module is capable of synchronous transfer of 16 independent channels of data.

4.8.2 Receive chain gain

MR receive signals cover a tremendous range, from near to the thermal noise floor (-174dBm/Hz) to -60dBm. The low end of the RXC dynamic range is determined by the RXC's noise figure (NF) performance. The upper end is determined by saturation effects and intermodulation distortion performance. Since it is difficult to optimize performance over such a large range, the receive chain has gain adjustments.

The RXC has two methods of adjusting receive path gain, which are called R1 (analog attenuation, pre-ADC) and R2 (post-ADC "digital attenuation").

4.8.2.1 R1

The *R1 gain control* adjusts the gain between the preamplifier and the digitizer (ADC).

The R1 setting controls an analog step attenuator, located after the preamps but before any frequency conversion stages. The attenuator has Is a 3dB/step, with minimum attenuation at R1=13, maximum attenuation at R1=1, for a total range of 39dB. The attenuator is on the RF switch boards (in the mixer module, in the RF Hub).

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R1 provides signal level control needed to prevent 1) saturation of the mixer inputs, and 2) over-range on the ADC input. The R1 setting is chosen during auto prescan, so that the signal level is 25-50% of the full-scale input value of the ADC. The manual prescan user interface has an R1 slider control.

System communication is thru the Hub via DVMR link. R1 can be changed during a scan.

4.8.2.2 R2

The *R2 gain control* adjusts the gain between the digitizer and the raw data storage. R2 exists to prevent data over-ranging in memory.

R2 "gain control" is implemented by bit shifting the digital data. Each R2 step is one bit shift, which is effectively 6dB. The bit shift takes place in the VRF (Volume Receivers and Filter) Module, which is part of the Recon Sub-System. The R2 value is set during auto prescan, so that the peak signal level is between 25-50% of the full scale value of the VRE register. The manual prescan UIF has an R2 slider control.

The value of R2 (for non-EDR, see below) ranges between R2 = 1 and R2 = 15. When the value of R2 = 14, the output of the ADC is unaffected. In other words, the rawdata put into memory reflects the same amplitude, relative to full-scale, as the input to the ADC. This means that ADC over-range and rawdata over-range will occur at the same signal level. At R2 = 15, the rawdata is 2X the ADC value (so anything above 50% on the ADC will result in rawdata overrange. For R2 values that are less than 14, the bit shift occurs to reduce the rawdata values, relative to the ADC values.

Only 16 bits are stored in rawdata (for non-EDR).

For multiple excitation scan (i.e. NEX > 1), the data from the repeated excitations is summed into the same registers; as 16-bit values are added into the same location, over-range can occur quickly. Therefore, the prescribed NEX value of a sequence will have an effect on the R2 value selected by prescan.

4.8.2.3 Signal resolution and "bit growth"

The output of the receiver's ADC is a 16-bit value, sampled at 80Msamples/sec. Since the MR signal bandwidth is much less, this oversampling can be used to increase the resolution (number of useful bits) of the data. This is referred to as "bit growth." Bit growth is occurs via digital filters in the receiver's FPGA; past input values are added such that the noise is filtered out and the signal is enhanced, resulting in about 18½ bits of valid and significant data. The value from the FPGA is sent to the VRE using a 32-bit word, therefore the VRE receives all 18-1/2 bits of significant data.

The VRF further filters the data; the filters reduce the data to the bandwidth chosen in the scan prescription. The output is stored as 50-bit register. The R2 value is an indication of which bits from within the 50 bits are kept and used in image reconstruction. For most scanning, 16-bit data is used; if EDR is selected (see below) 32-bit data is used.

4.8.2.4 Extended dynamic range

Since more than 16 bits of significant data may be available, Extended Dynamic Range (EDR) enables 32 bit rawdata; each I/Q pair consists of two 32-bit numbers, compared to two 16-bit numbers without EDR. EDR's 32-bit rawdata allows the system to take advantage of the process growth (aka "bit-growth") that occurs in the VRF.

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Enabling EDR generally doubles the memory requirements, non-EDR. For scans with many slices, multiple receive coils, and large rawdata matrix sizes, required memory grows very quickly. There is also some penalty in reconstruction speed with EDR. Therefore, if it does not make any difference in image quality, it does not make sense to enable EDR all the time.

EDR is enabled via an Image Option in the scan prescription. Some imaging sequences, such as 3D CUBE, have it permanently enabled, and will not allow it to be disabled. Low-noise (or "noise starved") scans need to retain all 18-1/2 bits of valid data from the ADC. For scans with more noise relative to the signal (like a 2D SE sequence), EDR is not needed.

4.8.3 Center frequency range

The system's RF center frequency (CF) will vary, based on multiple factors (magnet ramp and/or drift, siting considerations, and phantom chemistry, among others).

All RXC components must operate within spec over the following CF ranges:

For 1.5T: 63.860 MHz ± 64KHz

For 3.0T: 127.72 MHz ± 100KHz

4.8.4 Bandwidth

The RXC receive bandwidth can be adjusted, up to a maximum of ±250KHz.

4.8.5 Noise figure

Lower receive chain NF maximizes available SNR, which is more noticeable on low SNR images; there is no significant downside to lower NF.

"System" NF is usually refers to receive chain NF, measured from coil output to rawdata. The DV body path receive chain has an overall noise figure of <1.1 dB at R1=13.

Like most receive systems, system NF is strongly affected by preamp NF. The DV body preamp NF is <0.5 dB and typically ~0.3 dB.

Preamp NF is typically measured directly with a NF analyzer. System NF is usually measured by comparing image noise with a switchable noise source "on" and "off". This measurement is covered in B2.

4.8.5.1 NF vs. gain

NF depends on the RXC gain. Decreasing R1 adds RF attenuation before the ADC, and will degrade NF. Typically, NF<1.5dB at R1=13 (maximum system gain) and starts to degrade at roughly R1=7.

4.8.6 Maximum signal handling

Large signals will cause distortion in the received signals, due to saturation of amplifiers or mixers, or ADC overranges. Typical MR receive signal maximum amplitude is -50dBm at preamplifier input, and -20dBm at preamplifier output.

4.8.7 Transmit signal handling

The input to the receive chain is coupled to the transmit chain, either at the T/R switch or by the RF coil. While T/R protection and RF coil decoupling greatly reduce the amplitude of the transmit signal, it is still

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many orders of magnitude larger than a received MR signal. At the receive chain (preamp) input, transmit signals may be +20dBm (100mW) compared to a full scale receive signal of –50dBm (10nW).

The receive chain must be capable of withstanding this +20dBm signal level, without damage, and to recover from any saturation (caused by the transmit pulse) quickly (TDB) to allow MR signal acquisition soon after the end of the transmit pulse.

4.8.8 IM distortion

Intermodulation distortion (IMD) is refers to spurs created by the interaction of two signals (of slightly different frequencies) with a nonlinear system. A specific type of IMD, commonly used as an indicator of linearity, is the *third order intercept*.

The third-order intercept point (IP3) is based on a simple model for system nonlinearity: as the amplitude of the input signals increases, a third order spur will increase three times as much as the input. It can be modeled using a low-order polynomial, derived by means of Taylor series expansion. The third-order intercept point relates nonlinear products caused by the third-order nonlinear term to the linearly amplified signal.

The intercept point is a purely mathematical concept, and does not correspond to a practically occurring physical power level. In many cases, it lies far beyond the damage threshold of the device.

4.8.9 Single tone purity

4.8.10 T/R switching time

There are two aspects of TR switching related to the receive chain: multi-coil bias switching and receive chain blanking.

4.8.10.1 Multi-coil bias switching

The multi-coil (MC) bias drivers reside in the RF Hub. The RF Hub uses the transmit unblank signals (NB, BB and CW) it receives from the Exciter to control the MC Drivers. The MC drivers provide the bias signals for controlling PIN diode switches on the receive-only surface coils. They operate as programmable current source/sink circuits (generally during transmit) or voltage source circuits (generally during receive).

The PIN diode switches serve to make the surface coils transparent during transmit by "opening the coil loops". Generally, the PIN diode is biased on during transmit, enabling a resonant circuit; in turn, this prevents the current flow in the coil loops that would be induced by the RF transmit field. This "blocking" achieves two things. First, large circulating currents can lead to patient warming. Second, large circulating currents create a secondary B1 field that can interfere with the original transmit field and cause image quality problems.

The TR switching time depends on how fast the MC Drivers can switch states (either from transmit→receive or receive→transmit), settle to steady state and monitor for any fault conditions associated with their loads. In DV systems, the T/R switching time allotted for the RF Hub MC Drivers is 15 usec after the edge of the unblank signal.

4.8.10.2 Receive path blank/unblank

TR switching also controls the receive path blank/unblank. The receive chain is blanked during transmit for two main reasons:

- First, blanking provides protection from transmit-related signals for RXC components. The RF Hub blanks its circuitry since it connects directly to the coil and is therefore at risk. The Receiver Module blanks its input because the copper cabling connecting it to the RF Hub has the potential of carrying large voltages due to coupling to the transmit field.
- Second, it prevents invalid data, like large signals or transients induced by transmit, from propagating thru the receive path. Since the receiver's ADCs are always running, any noise entering the receiver would pass to the FPGA. By blanking (signaling the loopback switches to go into loopback mode) the ADC passes only zeroes to the FPGA during transmit. As added protection to prevent garbage data from entering the digital filters, the Receiver FPGA uses a Receiver Begin Acquisition (RBA) command from the PSD to begin the process of accepting data and filtering and ends when the view length has played out.

The receive unblank signal is controlled by the PSD, and passed to the RXChain via the DVMR link. The Receiver Module makes this control signal available to the RF Hub via the copper diagnostic and control path that connect the two components.

Both the RF Hub and Receiver Modules use this control signal to configure switches such that coil inputs are switched away from "thru paths" and into the switch's internal 50 ohm loads, protecting the downstream circuitry. This switch position is the same used in loopback mode when diagnostics or calibration is done. Since the receive unblank signal only controls switch blocks used to route the signals, they have the capability of switching between states very quickly. The DV system allows up to 7 usec to switch states, but switching speeds are typically less than 1 usec.

4.8.11 Multi channel isolation

Multi-channel receive capability requires multiple independent coils connected to the system through independent receive paths. It is important to keep these coils and channels isolated from each other to avoid coupling that can reduce image quality. The RXC should provide 30dB or higher isolation between channels. This can be measured to driving each channel with a signal slightly below full scale, and examining other channels for the signal's presence.

In the DV system, the coil generally dominates channel "cross-talk"; the receive chain is designed such that it does not significantly add to the cross-talk already present. Typically, coils have about 20 dB of isolation between channels. The RRx chain is designed for a minimum of 30 dB isolation, ensuring design margin.

Coupling or cross-talk between channels can come from connection points, circuit board layout and component choice and the type of cabling used to carry the receive signals.

- Individual connections from the coil (such as a single BNC connector) through the receive chain would offer the maximum isolation but this approach is not practical when there is a need for systems with 128-channel capability. The DV system utilizes multiple pin connectors with individual shielded coaxial cables to move the RF signals from the coil connection to the Hub.

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- Care is taken with trace layout, ground planes and component orientation on the boards within the Hub to maintain isolation between channels.
- After down-conversion to the Intermediate Frequency (16 MHz), copper-wire, twisted pair cabling is used between the Hub and the Receiver Module. Shielded twisted pair wire is used to improve the isolation in this path.

Once the receive signal enters the digital world (i.e. after the ADC in the Receiver Module) cross-talk between channels is a non-issue.

4.8.12 Multi-channel gain balance

Gain balance between receive channels is needed, so that 1) the final combined image intensity is more uniform across the whole image and 2) the signal to noise ratio of the individual channels is properly scaled. The gain from individual channels will not be exactly balanced due to several factors:

- Components and circuit variability, starting with the receive coil and continuing thru the receive chain (such as preamp gain, noise figure of the channels, etc.)
- Individual element or loop size differences and proximity to the anatomy of interest.

The RXC hardware has requirements placed on it to limit the gain variation between channels, from the coil port connector to the input of the ADC, to no more than 3 dB. In DV, all multicoil arrays have internal preamps that are outside this path. These preamps also have limitations placed on them regarding the acceptable variation in gain and noise figure.

4.8.12.1 Receive path calibration

In an effort to normalize the gain in the individual RRx chain pathways, a calibration (performed during TPS reset) measures and logs the different receive pathways. The log file is date and time stamped, and kept as `/usr/g/service/log/r1GainCal`. During this "R1" calibration, the signal level of each channel is measured in loopback mode while the subsystem steps through each value of R1 from 13 to 1. This calibration is necessary since, for a scan, all receive channels are set to the same R1 value; however the analog gain of each receive channel is not precisely the same for each R1 setting. The r1GainCal file contains the information for each channel regarding the delta from expected.

4.8.12.2 Prescan's noise calibration

Prescan contains a separate entry point for a "noise calibration" procedure. A noise frame is collected from all the active channels that will be used for the scan, and the noise signal's mean and standard deviation is computed. The goal of this process is to identify the relative gains between receiver channels for a given scan. The computed standard deviation values are called "prescan noise weights" and can be found in `/usr/g/service/log/paramdata/ prescanParameters.xml`.

This noise characterization pass uses the maximum R1 setting, but prescan's receiver gain (RG) pass will likely select a different R1 for image scanning. Changing R1 implies a change in relative channel gains; therefore, the computed prescan weights must be corrected accordingly using the receive path cals described above.

In addition to this correction, prescan applies a set of "coil weights" to the noise calibration. The coil weights are characteristic of the coil design and are found in the coil database.

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Reconstruction uses the final prescan receiver noise weights to correct each channel image. The process of noise weighting actually affects image quality and not just the way an image looks with respect to window and level. This is true because the weighting is actually applied within the algorithm used to reconstruct the combined image.

Since the Noise Calibration pass only works on the diagonal of the correlation matrix, it does not correct for variations in noise levels due to coupling between coil elements. It can, however, help in the case of a failing coil element where the signal is low and the noise is high.

4.8.13 Transient noise suppression

Transient noise (TN) events (also called spike noise or white pixels) are extremely wide band, very short duration bursts of electrical noise; typically, TN is caused by sudden electrical discharges or intermittent electrical conduction paths near the MR scanner. The emitted noise is picked up by the RF receive chain, and shows up as noise spikes in the received signal, as white pixels in the raw data, and as corduroy artifacts in the reconstructed images. (Hence the varied names applied to transient noise.)

While the level of TN can be reduced by hardware design, it is not always practical to eliminate all TN sources. Therefore, the receive chain includes mechanisms to detect TN, and minimize the effects on the reconstructed images.

Because TN events have energy in the same bandwidth as MR signals, they cannot be directly filtered out. However, if the time location of an event is known, the receive signal can be blanked to suppress the noise spike; because the TN signal has wide bandwidth, it can be effectively suppressed by a very short blanking pulse (on the order of 10-100ns). Because the MR signal has much lower bandwidth, it varies slowly in comparison, so the effects of a short blanking pulse are negligible.

4.8.13.1 TN event detection

Transient noise events and the MR signal have different frequency content; the TN signal is very wideband, limited by the bandwidth of the receive chain, while the MR signal is much narrower bandwidth. This difference can be used to discriminate between TN signals and MR signals for reliable event detection.

A wide bandwidth copy of the received signal, with a mix or MR and TN signals, is filtered to block the MR signal. The resulting TN only signal is amplitude detected; because it is wideband, the TN event is very narrow in time and it's time location can be accurately determined.

Typically, the signal filtering of the TN detection path is fixed. The detection threshold is programmable under system sequence control, and is typically set to around ten times the rms noise voltage at the detection point. (needs to be checked.)

4.8.13.2 TN event suppression

Several workable methods for reducing the effects of TN are available. Both apply a short blanking pulse to the MR signal, either via an analog switch or thru digital signal processing.

4.8.13.2.1 RF signal path blanking

An "analog" method for TN suppression uses a high speed switch to blank the receive signal when a TN event is detected. Because the switch has finite response time, a delay line is used to align the receive signal's

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timing with the switch response. The switch must be designed carefully to avoid introducing switching noise or additional transient signals into the received signal.

The blanking pulse duration is programmable, under system sequence control.

4.8.13.2.2 Digital blanking

If the digitized signal has sufficient bandwidth, digital blanking can be used. The samples contaminated by TN can be replaced by zeros, or by interpolated samples. This eliminates the problem of the high speed switch design.

In a digital suppression implementation, both the blanking delay and blanking duration can be programmable under system sequence control.

4.8.14 Receive signal routing

RF signal routing covers switching or routing of received MR signals. Typically, the system has more receive signal ports than actual receive channels. Multiple user accessible connectors are provided to simplify coil connection, provide for different coil connectors, or other reasons.

An RF signal routing system selects which of the multiple inputs is routed to the receivers in the receive chain. The routing is done after preamplification, preventing loss of signal quality without requiring low noise figure input on the routing.

For multiple coil receive paths, "bank selectable" routing is used, so signals are routed as groups of eight or sixteen. Channel by channel control is not needed. This simplifies control and prevents unintentional channel swaps.

4.8.14.1 DV routing implementation

While a DV system is capable of supporting 128 independent receive channels, a typical DV system has 32 independent receivers.

The DV system can have many as five connection "ports" for receive only coils; each port can handle multiple receive signals. These ports are designated Connector A (which also has transmit capability), P1 (also with transmit capability) and P2, P3, and P4. They are located at both ends of the patient table for user convenience. Connector A has eight independent receive channel capability, while all of the P-connectors are capable of interfacing with 32 independent receive channels.

Since there are generally more coil connection points than receiver channels, it is necessary to have routing or multiplexing capability to connect an active channel to an available receiver. In DV systems, the routing of available coil ports to available receivers is done in the RF Hub assembly. This includes routing for both the body coil signals and local transmit coil signals.

The routing of coil receive paths is accomplished on the RF Switch Board within the Hub. The RF Hub also contains a Control Board that receives the information from the system regarding proper configuration for a scan; including the routing required as well as the proper setup of the multi-coil drivers. This information is passed to the Control Board during scan setup via a CAN (Controller Area Network) communication link. The CAN communication link uses a DSP and is therefore disabled during the actual scan to prevent noise. Any

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real-time communication that is needed by the Hub during the scan (such as RRX unblank or Switch on the Fly information) is passed to the Hub from the Receiver Module via a copper communication path.

Routing of the channels is done in four contiguous RF signals in banks of 4 to 4 outputs; for a total of 16 outputs per receiver. The individual RF input channels are grouped into banks of four on each input connector on the Switch Board. The combined 16-channel output connector from the RF Switch Board to the Receiver is separated into four banks of four for routing purposes. The surface coil designer must have knowledge of the routing scheme such that they properly connect the coil elements to the port connector pins to achieve the desired groups of coils. This is especially important in multicoil arrays with a large number of elements such as the GEM PA located inside the table and is less important for smaller channel count coils such as the HR Brain coil where all 8 elements are always in use simultaneously.

4.8.14.2 Switch-on-the-fly

The DV system has a "switch on the fly" feature that allows different channel sets of a multicoil to be used within the same scan. This feature is useful when performing exams such as peripheral vascular studies, where it is desirable to stitch together images from the pelvis area to the patient's feet.

This feature allows the user to choose between 1) using separate prescan passes for each station prior to starting the scan, or 2) completing only one prescan pass on the first station and re-using all the parameters for successive stations.

With Switch on the Fly, the RF Hub is sent all the pertinent information for setup of the entire scan and then changes its setup parameters during the scan based on a write signal passed to it from the Receiver on the copper link. The Hub is capable of changing its routing scheme and R1 gain, if necessary, within 7 usec of the command activation.

4.8.14.3 Biasing with multiple coils connected

It is possible to connect multiple MC arrays to the system simultaneously and change the receive coil of interest during the exam without re-positioning the patient. All unused coils that are connected to the system are provided with a constant transmit bias to keep them "blocked" or turned off while other coils are being used.

4.8.15 Receive signal processing

The main function of the RXC is to accept the RF signal from the receive coil (~64 MHz at 1.5T and ~128 MHz at 3T) and convert it to a digital signal that can be utilized by the recon sub-system. In the DV system the signal processing applied to each channel includes 1) down-conversion or frequency conversion, 2) analog to digital conversion, and 3) filtering and decimation.

4.8.15.1 Frequency conversion

The implementation of DV converts the single-ended RF signal (coming from the coil port at the Larmor frequency) to a differential signal at a 16MHz intermediate frequency. This makes the IF signal independent of the source: it may be from a 1.5T system, a 3T system, or a non-1H spectroscopy signal. The receiver does not need to change. Down-converting to a 16 MHz differential signal allows cabling between the RF Hub and the Receiver Module can to use twisted pairs and sub-D connectors (as opposed to coaxial connectors and cabling.)

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The signal conversion takes place within a Mixer Module that resides on the RF Hub Switch Board. The frequency conversion is accomplished by mixing (or multiplying) the 64 MHz or 128 MHz signal with a Local Oscillator (LO) signal to generate a signal at nominally 16 MHz. The value of the IF frequency was (carefully!) chosen by considering possible harmonics generated by the combination of the IF, the LO, and the possible center frequency of the system. 1.5T uses an 80MHz LO; at 3T it is 112 MHz. The LO is generated in the DV Exciter and is sent to the RF Hub via a twinax cable as a differential signal to minimize noise susceptibility.

4.8.15.2 A/D conversion

The Receiver Module contains a high quality analog to digital converter (ADC) for each of the 16 channels of data. The ADC has a resolution of 16 bits and a sampling frequency of 80 Msamples per second (Msps).

The input to the ADC is the 16 MHz data from the RF Hub. The full-scale input range of the converter is 3.2 Vp-p across its differential input or 1.6 Vp-p on each input. The output data format is twos complement and the Most Significant Bit (MSB) is the sign bit, which results in a full scale digital range at the ADC output of -32768 to 32767.

The MPS R1% is the true "percentage of full-scale value" at the ADC and comes from the receive channel with the greatest signal. The R1% value is important because it is critical that the ADC does not over-range during the scan.

An over-range has an obvious negative affect on image quality. The over-range detection of the ADC sampled data is done in the FPGA.

The ADC will reach full scale when the Receiver Module input is 4±1 dBm.

4.8.15.3 Digital filtering (to be refined)

The receiver module's FPGA receives the 16-bit data from the ADC. The FPGA provides quadrature demodulation, filtering and decimation functionality. The MR signal that is digitized by the ADC has a bandwidth of +/- 250 kHz. As indicated above, the ADC samples this data at a rate of 80 MHz. Since we have so many samples at the output of the ADC we can decimate the data (reduce its size) without losing any information. We use all of the samples in an intelligent way to derive one sample through the use of digital filters. This decimation process is not the same things as simply throwing out all but every nth sample. If this were done, it would be no different than reducing the ADC sampling rate drastically and we'd lose the benefit of the 80 Msps rate.

An early process performed in the FPGA is quadrature demodulation that lowers the center frequency to baseband (0 Hz). Because it is tuned with a quad oscillator it retains amplitude and phase information and there are two outputs (I and Q) from this block. These two outputs remain separate from this point through reconstruction.

The data is then processed through CIC (Cascaded Integer Comb) filters and FIR (Finite Impulse response) filters. These filters use past input data over a rolling window to decimate the data. The past inputs are added with knowledge of the time delay through the circuit and desired output bandwidth to produce an output where the noise is filtered out but the signal is enhanced. This process is also known as Process Gain or Bit Growth. Because of this filtering and decimating process the 16-bit input into the FPGA becomes 18-1/2 bits of valid data at the FPGA output with a bandwidth of +/- 250 kHz. This is the data that is sent over

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the DVMR Link (Fiber Optic connection) to the VRF (Volume Receive Filter) which is part of the Recon sub-system.

4.8.16 Further information

4.8.16.1 Preamplification

The preamplifier is critical to overall receive chain performance, as it is at the front end of the receive chain. Desired performance includes

1) low noise figure
2) high gain
3) low input impedance
4) stability over all input load impedances
5) tolerance to large input signals (from RF transmit)

Note that preamplifiers may be integrated into RF coils in many systems. However, they are considered part of the receive chain.

4.8.16.1.1 Low input impedance

The preamp's input impedance is used as a critical decoupling network circuit element for multi-coil designs. Most designs implement a low input impedance, using a series resonant charactistic with manual tuning to adjust phase to zero. Typically, a tradeoff must be made between low input impedance and the preamp's physical size, as the high Q components needed for the resonant circuit are physically large.

Note: High input reflection coefficient (|S11|) is actually the critical performance parameter, as opposed to low input impedance. Transformers or controlled wavelength cables can be used to shift S11 phase to zero.

Typical input reflection coefficients are <0.3dB. (needs to be checked)

4.8.16.1.2 Stability over all input impedance loads

Preamps are tightly coupled to the RF coil. Because the output impedance of the RF coil varies with loading, the preamps must be stable over the expected RF coil impedance range.

4.9 RF coils

This section covers the design approach and rationale for GEHC MR Premium systems RF coil design.

The content below uses common RF coil terms (transmit vs. receive, volume vs. local, quadrature, coverage, multiple channels, etc.) that are covered in section 3.8.3 *RF coils*.

4.9.1 General design approach

The GEHC MR Premium systems share a common RF coil design approach, although the implementation varies across different products.

1) For transmit:

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a) The body coil (BC) is used on medium to large FOV (greater than about 16cm) applications. The body coil is built onto the bore wall former, and is not customer removable. Quadrature transmit is always used, to minimize SAR.

b) A head volume transmit coil *may* be provided, for use in neuro applications, and often for neonatal applications. If the head volume transmit coil is not available, body coil transmit is used for head and neck applications. It has been difficult to overcome the transmit field uniformity limitations of head specific transmit coils, particularly in the neck and upper shoulder regions.

c) Specialized removable volume transmit coils are used on some small FOV applications, like knees and wrists. Quadrature transmit is used.

d) Local transmit coils are *not* used, or are limited to specialized use, as the SAR controls required by local transmit coils are not built into the system.

2) For receive:

a) Multi-channel coils are almost exclusively used, for the advantage of high SNR.

b) A multi-channel receive *posterior array* coil (the PA) for spine and torso imaging is usually built into the system's patient table. Typically, this coil has at least 32 coils. Multiple configurations, which use different groups of coils, can be chosen by the operator. This coil is generally not removable, although tables without a coil installed may be used.

c) A multi-channel receive array coil for head, neck, and upper spine (the HNU or HNS) is available and widely used. The coil supports multiple configurations, which use different groups of coils, selectable by the operator. This coil is removable.

d) Many specialized multi-channel receive coils are available. Most are targeted to a region of the body, like a shoulder or pelvis. In the case of wrist and knee coils, they may be combined into the same assembly with the volume transmit coil.

3) Coil connectors:

a) The body coil and posterior array have no customer accessible connections, as they are not removable.

b) Several connection points for removable coils are located on or near the patient cradle. These connectors include multiple RF receive channels, enable/disable biasing, and lines to identify and check for presence of the coils. At least one connector will also support RF transmit power for use with small volume transmit coils.

4.9.2 Quadrature

Almost all volume coils are designed for "quadrature", both on transmit and receive. The MR signal's precession inherently generates a quadrature signal, from the precession around B0. Quadrature coils take advantage of this; linear coils use only a portion of the signal.

On the transmit side, a quadrature transmit coil requires half the power of a linear coil. This factor-of-two reduction has two implications. First, more imaging can be accomplished within allowable SAR limits.

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Second, less expensive transmit chain components (smaller RF amps, smaller cables, etc.) can be used. All major high field systems (including all DV systems) use quadrature transmit body coils. Most implementations use birdcage coils with two drive points, physically located with 90 degree separation, and driven with quadrature inputs (equal magnitude and 90° phase shift).

On the receive side, coils can be designed with at least two loops that are placed orthogonal to each other, so that the coil picks us both components of the quadrature signal. The signal from the two loops will be correlated, while the noise will not, so proper combination will yield a root 2 advantage in SNR (41% increase, compared to linear).

4.9.3 Birdcage coils

Birdcage coils are widely used throughout the MRI industry, for both transmit and receive. The most common example is the body coil.

Birdcage coils are cylindrical volume coils, with the axis of the cylinder aligned to B0. This makes them ideal for cylindrical MRI systems. The can create highly homogenous transmit and receive fields. The "birdcage" moniker due to the look of the physical structure; there are two end-rings, connected with several (typically 8, 16 or 32) runs. The rings and runs are formed of copper traces and may contain lumped elements.

4.9.4 Channels

RF coils can be categorized by channel count. In early systems, only single-channel surface coils existed. The early-1990s saw a revolution in coil design with the introduction of multiple channel receive coils, often called "Multi-Coil" coils. Multiple channel coils have become essential to high performance imaging, and now dominate clinical MRI practice.

Single channel coils are typically 1) a single loop, or 2) two loops whose signals were combined on the coil into as a single output to the system.

In a multi channel coil, multiple independent and over-lapping coil elements are contained within one physical structure. Each element has its own receive path, and produces an independent receive signal. These separate signals are combined in the reconstruction process into a combined image. The main advantage to coil arrays is the increase in image volume that can be simultaneously acquired while maintaining the SNR of a smaller, single loop coil. The independent coil elements within an array are over-lapped to improve isolation between adjacent loops.

4.9.5 "Metacoils"

A metacoil is defined as a combination comprised of multiple physical coils, each using a different interface port. As an example, the GEM 3T Neck 36 (coil mode) uses two physical coils: The GEM PA, plugged into P3 and the HNU Face, plugged into P1. These two coils are defined as compatible to be in the bore simultaneously and only a sub-set of each physical coil's elements are used as receive coils in this configuration.

4.9.6 Decoupling Receive-Only Coils

Receive-only coils must be *decoupled* (aka. "turned off" or "blocked") during transmit pulses. If the "unblocked" receive coil (a tuned loop) is inside the transmit volume, Faraday's Law says that the B1 transmit field would induce large circulating currents in the receive coil. These currents would cause two problems.

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First, they would generate their own field, opposing and causing distortion to the B1 field, which in turn causes image quality problems. Second, the circulating currents generate heat in the receive-only coil, which impacts patient safety. Therefore, all receive-only coils have some decoupling mechanism for use during the transmit pulses.

These decoupling networks are commonly made up of resonant L-C circuits with an active PIN diode switch. The standard approach is to bias the PIN on (with and external *bias driver*) during transmit, which enables a parallel LC resonance to block RF currents – it "breaks the loop open."

Some coils use "passive" decoupling. In this circuit, two "crossed" diodes are placed in opposite directions; the voltage induced by the B1 transmit field turns them on. These circuits do not require external DC bias to function.

Some coils use both types of decoupling circuits.

4.9.7 RF Coil Design Considerations

4.9.7.1 Magnetic fields

As described above, the B1 transmit field induces large voltages on RF coils placed within the transmit field. In turn, this may lead to heating of the coil elements, blocking networks, and coil cabling. The coils must not only survive these stresses, but must stay within allowable temperature limits. In general, the *peak B1* strength determines the maximum voltage stress on the RF coils, while the RMS of the B1 (often shortened to *B1rms*) determines the maximum heat stress on the coils. The system safety limits (see section 4.2.2.1) include limits on peak B1 and B1rms specifically for this reason.

RF coil designs must be tested to ensure that they can tolerate these stresses. Since most surface coils can be positioned within the bore in many locations, the coil must stay within the surface temperature heat limits for all possible configurations. The testing to ensure this is a large part of the surface coil design effort.

Coil designs must also not significantly distort the B1 transmit field. A surface coil is considered "transparent" if the B1 transmit field is the same with and without the surface coil present.

Surface coils must also be designed to withstand, without damage, the magnetic field strength produced by the gradients. Because gradients operate at have much lower frequencies than RF (by a factor of 1000), this is generally not a significant design challenge.

4.9.7.2 Coil ID

The system operator will use many different RF coils during a typical day, placing and removing them with each patient. User-removable RF coils (all coils, with the exception of the body coil) must have the capability to identify themselves to the system– this feature is called Coil ID. Each coil design is given a unique identification value that is read by the system. This information is used by the system to properly configure the system setup, and to prevent incompatible coils sets from being placed in the bore simultaneously.

Coils that have multiple hardware configurations, such as detachable elements, must have the capability to identify the various configurations.

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4.9.7.3 Materials

The materials used in surface coils 1) should be non-magnetic to avoid distorting the B0 field, and 2) should not produce any MR signal.

Magnetic materials cause two problems. First, even small amounts of materials may cause enough B0 distortion to affect image quality. The standard test is to place a material sample near a phantom and look for B0 distortion effects with a sensitive imaging sequence. Second, large amount of magnetic materials would cause the RF coil to be attracted to the bore, which is a physical hazard. In general, any materials or part that exhibit detectable pull are not allowed in RF coil designs.

Materials should also be "MR silent." Most solids have much shorter T1 and T2 times compared to human tissue. Therefore, very short TE sequences are used to test for MR signals from candidate materials. As an example, the introduction of RUFIS in DV24 reduced the TE time enough that the housing of the GEM HNU coil can now be seen in the image. While the resulting artifact is outside of the imaging volume of interest so it does not interfere with the diagnostic quality of the image, the ongoing push toward shorter TEs (like with RUFIS) should be considered with future material choices.

4.9.7.4 System compatibility

Coils must be designed to be compatible with the system interface.

In DV systems, this interface is thru the RF Hub, which has specific multi-coil bias capabilities for transmit decoupling as well as receive channel routing. The multicoil drivers in the Hub provide a constant current source during transmit (surface coil decoupled) and a constant voltage source during receive (surface coil enabled). The elements within the array must be connected to the system such that logical groups of coils can be selected to provide maximum flexibility when using subsets of elements within a single array.

4.9.7.5 Patient related considerations

Coils must be designed to preclude patient safety or patient comfort issues. Considerations include pinch points for the patient or operator, abrasive or bio-reactive surfaces, cables that route over the patient during scanning, or preventing or interfering with emergency egress of the patient.

4.9.7.6 Coil Database (CoilDB):

The Coil Database was introduced in DV to replace the coil configuration files used in previous platforms. The database contains all the coil configuration information; the coil configuration is a set of entries related to coil physical properties, coil safety parameters, coil modes selection and prescan and imaging parameters. ***Physical properties***: the coil database defines physical properties such as the coil identification value, the current and voltage levels needed to decouple the coil, the pins used to connect to the system and the allowable maximum B1 RMS value for the coil.

Coil Modes: Each coil database record actually defines a specific *coil mode*, as opposed describing a specific physical coil. A physical coil array may have several modes of operation (different ways of using the coil). Each mode would have a separate database record. Examples include:

1. HNU coil: the "Head24" coil mode uses only a sub-set of the HNU's elements, while "Head+Neck40" is a coil mode that uses all 19 HNU elements.

2. The GEM Posterior Array (PA), which is built into the flat GEM table, is another example of a single array that has many coil modes. The PA coil has over 30 separate, individual elements and there are over 20 separate coil modes developed for this coil. Each mode uses different element sub-sets, depending on which area of the body is being scanned.

Having separate entries for each coil *mode* allows definition of the "logical" properties for that particular mode: for example, the MC bias/receive pin selections needed to activate a particular set of elements.

SAR characteristics: If the coil is a transmit/receive coil the database defines several safety parameters related to the SAR curves applicable to the coil.

User related information: The database contains information related to the coil's intended application and use, including

1. the clinical name the user will see on the user interface,
2. the generic group the coil will be associated with (such as spine, head, chest/torso/pelvis, etc.) and
3. Image scaling factors associated with the coil such as for PURE (Multi Channel RF Coil Sensitivity Normalization) or SCIC (Surface Coil Intensity Correction) values.

Prescan: The coil database is used by Prescan to retrieve parameter values Prescan uses for a given coil.

For further information on the coil database reference DOC0775678, Coil Data Interface Standard.

4.10 Patient handling

The patient handling (PH) subsystem is designed to position the patient properly in the magnet bore, keep the patients and operators safe and comfortable, protect the system components, provide displays and controls for the operator, and create a pleasing aesthetic look to the system.

Major PH components include:

1. Operator interface
2. Patient interface
3. Patient transport
4. Magnet enclosures

Safety design considerations for the PH subsystem as a whole are listed below.

1. Protection from high voltage: Both the RF transmit coils and gradient coils have hazardous voltages present when operating. The PH subsystem design protects both patient and operator from these voltages.
2. Protection from mechanical hazards: The PH subsystem design protects both operator and patient from pinch points and crushing hazards.
3. Protection from fire hazards: The PH subsystem design includes the use of non-flammable or flame-retardant material to minimize risk of injury due to fire.
4. Protection from bio-hazards: The PH subsystem design includes the use of non-bioreactive materials on exposed surfaces. This material also is able to be cleaned and disinfected.